



CHAPTER I

Automated Gas Refueling Management System with Cashless Operated Machine for RS GAS

A Capstone Project

Presented to the Faculty of

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Submitted Date:

Abstract

This manuscript outlines the development and implementation of an Automated Gas Refueling Management System featuring a Cashless-operated Machine for RS Gas. The primary goal of this project is to enhance customer convenience, simplify payment processes, and ensure secure cashless transactions. By integrating modern technology into the gas refueling process, this study addresses key issues related to customer experience, operational efficiency, and security. The findings indicate that the proposed system significantly improves the gas refueling experience for customers while optimizing operations for RS Gas, positioning the company as a leader in innovation within the fuel industry.



PROBLEM AND ITS BACKGROUND

This chapter provides an overview of the study, including its background, theoretical framework, conceptual framework, statement of the problem, scope and delimitations, significance of the study, and definition of terms used. It sets the foundation for the research by outlining its key components and objectives.

Introduction

The fuel service industry in the Philippines remains a critical component of the transportation and logistics sectors, supporting both private and commercial mobility. However, traditional gas stations in the country still rely heavily on manual processes and cash-based transactions, which often result in long queues, operational inefficiencies, and customer dissatisfaction. These outdated systems are increasingly incompatible with the expectations of today's consumers, who demand faster, safer, and more convenient services driven by digital technology.



RS Gas, a local fuel provider, recognizes the need for innovation in order to stay relevant and competitive. In response to evolving consumer preferences and the ongoing digital transformation of service industries, RS Gas seeks to implement a modern, automated gas refueling management system that integrates cashless payment technologies. This system is designed not only to reduce human error and operational delays but also to enhance the overall customer experience through secure, self-service refueling stations.

Globally, fuel providers are adopting automation and digital payment systems to meet growing demands for speed and security. In the Philippines, similar changes are beginning to emerge, but smaller or independent stations often struggle to keep pace. By adopting an automated and cashless-operated machine, RS Gas aims to lead the way in local fuel service innovation, providing a solution that improves customer service, ensures transactional safety, and streamlines operations.

This study explores the impact of implementing such a system, particularly its potential to transform customer interactions, reduce wait times, and boost operational effectiveness. Moreover, it investigates user perception and readiness to adopt cashless technologies in everyday refueling activities. As digital infrastructure continues to expand across the Philippines, the shift toward automation in service delivery becomes not only viable but essential.



The study will also assess technical and operational challenges involved in the development and deployment of the system. Through this, the project aims to provide valuable insights that can support future improvements in similar service industries. Hence, the central research question guiding this study is: How can an automated, cashless-operated gas refueling system improve customer satisfaction and operational efficiency for RS Gas?

Background of the Study

The traditional method of gas refueling in the Philippines remains largely manual and reliant on human attendants and cash payments. While this setup has served its purpose for decades, it now poses increasing challenges in an era where digital transformation and automation are rapidly redefining consumer expectations. Issues such as long service times, limited transparency, and the risk of cash mishandling highlight the need for a more modern solution.

With the rise of digital payment platforms and the Philippine government's push toward a "cash-lite" economy, businesses across sectors are being encouraged to adopt cashless solutions to enhance convenience and security. Within the fuel industry, however, only a handful of large stations have begun this transition, leaving many independent gas providers at a technological disadvantage.

RS Gas sees this as both a challenge and an opportunity. By pioneering an Automated Gas Refueling Management System with a Cashless-operated Machine, RS



Gas intends to revolutionize its service model. The proposed system allows customers to independently operate a refueling machine, make cashless payments through cards or mobile wallets, and minimize contact during transactions, a feature particularly relevant in a post-pandemic world.

This transition also presents a strategic business move that aligns with consumer behavior trends and industry benchmarks. Through improved speed, accuracy, and customer autonomy, RS Gas can not only elevate its service standards but also reduce the workload on human staff and minimize operational costs in the long run.

The study therefore examines not only the technical feasibility and user satisfaction of the system but also its implications for broader industry transformation. By developing and testing this solution, the project aims to contribute to the modernization of fuel service operations in the Philippines, setting a potential model for similar enterprises.

Theoretical Framework

Technology Acceptance Model (TAM), developed by Fred Davis in 1989, serves as a foundational theory to understand how users come to accept and use technology. Rooted in the Theory of Reasoned Action, TAM proposes that two core beliefs, Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) determine an



individual's intention to use a system, which subsequently influences actual usage behavior. Perceived usefulness refers to the degree to which a user believes that using a specific technology will enhance their job performance, while perceived ease of use reflects the degree to which the user expects the system to be free from effort. These two factors shape users' attitudes toward a new system, ultimately predicting whether the system will be adopted or rejected.

The model illustrates that external variables, such as system design, user interface, and functionality, impact perceived ease of use and usefulness, which then influence users' attitudes, intentions, and actual behavior. Notably, perceived ease of use indirectly affects system use by influencing perceived usefulness. The model highlights that even if a system is considered useful, users are less likely to adopt it if it is too complex or difficult to navigate. This balance between expected benefit and ease of interaction forms the psychological basis for user acceptance.

In the context of the Automated Gas Refueling Management System with Cashless-operated Machine for RS Gas, the Technology Acceptance Model is especially relevant. As RS Gas transitions from traditional, manual refueling processes to an automated, cashless solution, understanding user behavior is critical to the system's success. TAM provides a framework for evaluating whether customers perceive the new system as beneficial in reducing time and effort, and whether they find it easy to use. The adoption of RFID tags, contactless payments, and a user-friendly interface must therefore be aligned with these two core constructs of TAM.

By applying TAM, the developers can pre-emptively identify barriers to user acceptance and optimize the system accordingly whether by simplifying interfaces, enhancing functionality, or offering user education. The theory not only aids in system design and evaluation but also supports long-term strategic decisions regarding the integration of future digital features. In doing so, RS Gas strengthens its technological competitiveness and ensures that its innovations are not only advanced but also embraced by its customers. This alignment of technology design with user behavior solidifies TAM as a valuable theoretical guide in modern fuel automation solutions.

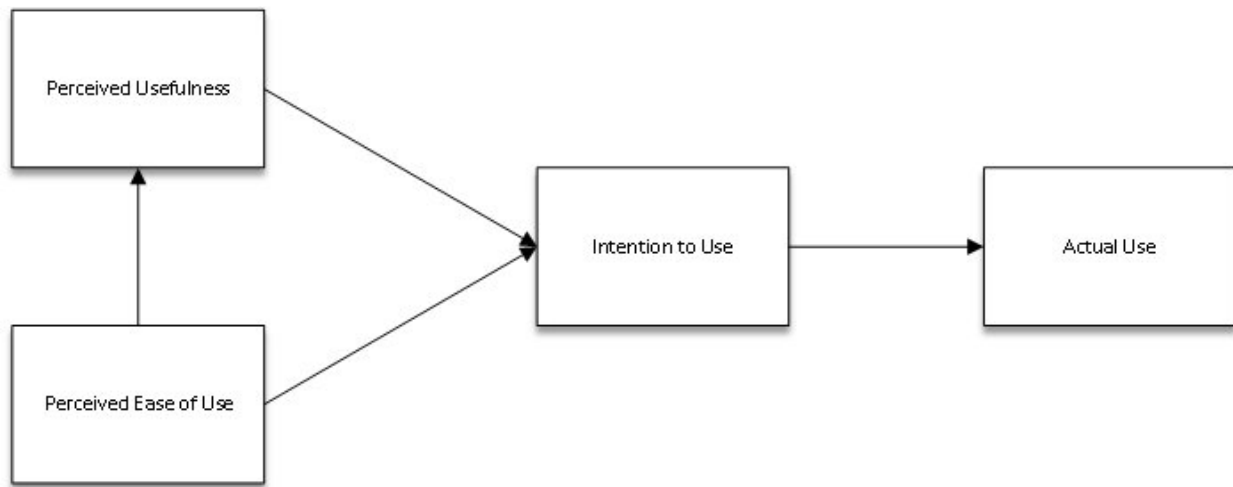


Figure 1: Technology Acceptance Model (TAM) Theory

Conceptual Framework



The proposed conceptual framework aims to evaluate the impact of implementing an automated, cashless gas refueling system on consumer satisfaction and operational efficiency at RS Gas stations. The framework begins by identifying key input factors, particularly the demographic profile of respondents (age, gender, frequency of gas refueling, and payment method preference), which help segment and understand target customer behavior.

The process phase utilizes a quantitative research approach, involving surveying and questionnaires to collect primary data from consumers. These are followed by data interpretation and analysis of consumer perception and behavior toward the proposed automated system. Key focus areas include perceived convenience, transaction speed, payment security, trust, and willingness to adopt a cashless payment method.

In the output phase, the framework aims to assess the effect of cashless-operated machines on customer satisfaction and wait times, providing insights into system performance and consumer response. Additionally, it identifies recommendations for improving the automated system, based on user feedback and observed operational challenges. The goal is to deliver practical strategies that can enhance user experience, streamline operations, and increase adoption rates of automated services at RS Gas stations.

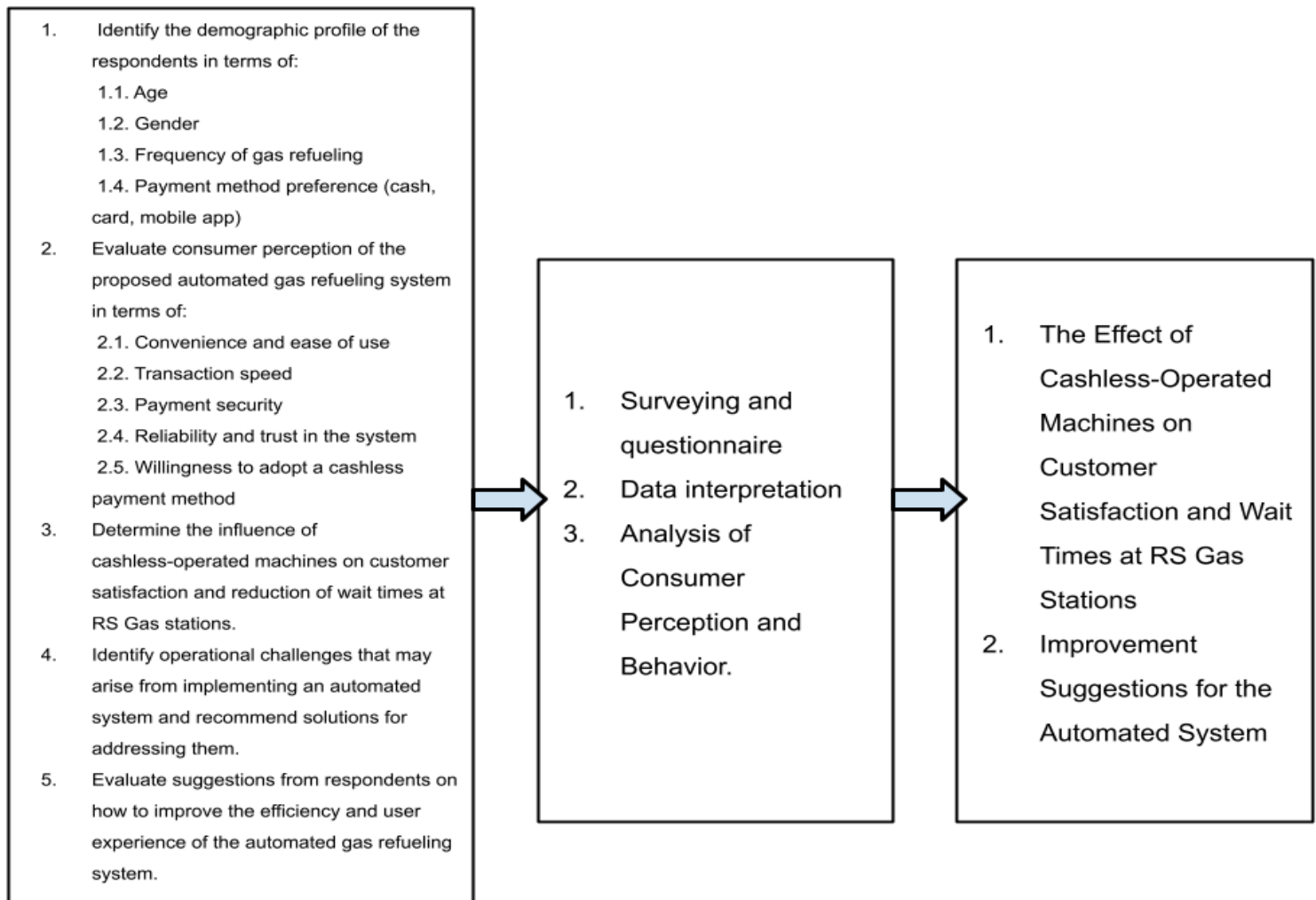


Figure 2. Conceptual Framework Paradigm



Statement of the Problem

The current gas refueling process at RS Gas relies heavily on manual operations and cash-based transactions, which often result in long wait times, human error, and security risks. As customer expectations shift towards faster, more secure, and convenient digital experiences, there is a pressing need for RS Gas to innovate its services. This study aims to assess the potential impact of an Automated Gas Refueling Management System with a Cashless-operated Machine on customer satisfaction, operational efficiency, and transaction security.

Specifically, this study seeks to answer the following questions:

1. What is the demographic profile of the respondents in terms of:

1.1. Age

1.2. Gender

1.3. Type of vehicle owned



1.4. Frequency of gas refueling

1.5. Payment method preference (cash, card, mobile app)

2. How do consumers perceive the proposed automated gas refueling system in terms of:

2.1. Convenience and ease of use

2.2. Transaction speed

2.3. Payment security

2.4. Reliability and trust in the system

2.5. Willingness to adopt a cashless payment method

3. Does the implementation of a cashless-operated machine influence customer satisfaction and reduce wait times at RS Gas stations?

4. What operational challenges may arise in the implementation of an automated system, and how can they be addressed?

5. What improvements do respondents suggest for enhancing the efficiency and user experience of the automated gas refueling system?

By addressing these questions, this study aims to provide insights into how RS Gas can enhance its service offerings to meet customer expectations in the context of evolving technology and increasing demand for digital and cashless solutions. The results of this study will serve as a foundation for potential improvements in customer experience,



operational efficiency, and system security, ultimately helping RS Gas stay competitive in the evolving fuel retail industry.

Significance of the Study

The growing demand for digital transformation in the fuel service industry has highlighted the limitations of traditional, manual gas refueling systems in the Philippines. Long queues, cash handling risks, and inefficient service delivery have made it necessary for providers like RS Gas to explore automation and cashless technologies. This study is significant in addressing how an Automated Gas Refueling Management System with a Cashless-operated Machine can improve service quality, operational efficiency, and customer satisfaction in line with modern technological standards.

To the RS Gas Management

The study provides valuable insights into customer behavior, preferences, and responses to digital innovations. It will assist RS Gas in evaluating the effectiveness of an automated, cashless-operated machine in reducing wait times and improving service efficiency. The findings can help management implement targeted strategies for smoother operations, enhanced customer service, and increased competitiveness in a tech-driven market.

To the Policymakers

This research supports national efforts toward a “cash-lite” economy by showcasing



how cashless technologies can be effectively implemented in essential services like fuel refueling. The results may help policymakers design supportive frameworks or incentives for the adoption of automated systems, ensuring safer, faster, and more transparent service delivery in the public and private sectors.

To the Investors

Investors will benefit from understanding how RS Gas is positioning itself through technological innovation. The study offers data-driven insights into potential returns, cost efficiencies, and customer loyalty that automation can bring informing investment decisions in the fuel service sector's modernization.

To the Future Researchers

The research can serve as a foundational reference for those studying digital transformation in traditional industries. It provides a real-world case on how automation and cashless systems affect consumer behavior, operational challenges, and service innovation, paving the way for further exploration in technology acceptance and service design.

Scope and Delimitation of the Study

This study, titled *"Automated Gas Refueling Management System with Cashless-Operated Machine for RS Gas,"* focuses on evaluating the impact of



implementing an automated, cashless refueling system at selected RS Gas stations around Luzon. It aims to assess the system's effectiveness in enhancing customer satisfaction, reducing wait times, and improving operational efficiency. The study also explores consumer perceptions regarding convenience, transaction speed, payment security, and willingness to adopt cashless methods.

The research will involve data collection through surveys and questionnaires distributed to RS Gas customers and staff at selected branches across Luzon. A manageable sample size will be used to gather focused insights, allowing for detailed analysis of user experience and operational challenges.

However, this study is limited to RS Gas branches within Luzon and does not include data from stations in Visayas or Mindanao. It also excludes comparisons with other fuel companies and does not aim to generalize findings across the entire fuel industry. The technical aspects of system design and backend integration will not be covered in depth, as the focus remains on customer-facing performance and adoption.

Definition of Terms

To ensure clarity and a shared understanding of the concepts discussed in this study, the following key terms are defined as they are used in the context of the research:



Automated Gas Refueling Management System

Refers to a technology-driven system that allows customers to refuel their vehicles without the assistance of station personnel. It includes self-service fuel pumps and integrated software for monitoring, managing, and recording refueling transactions.

Cashless-Operated Machine

A payment system integrated into the gas refueling machine that accepts non-cash transactions such as debit/credit cards, mobile wallets (e.g., GCash, Maya), and RFID-based payments, allowing customers to pay securely and efficiently without physical cash.

RS Gas

A local fuel provider in the Philippines, particularly operating across Luzon, which serves as the case subject of this study and the primary beneficiary of the proposed automated refueling and cashless system.

Customer Satisfaction

A measure of how products or services meet or exceed customer expectations. In this study, it refers to the customers' level of satisfaction with the automated and cashless refueling experience at RS Gas stations.

Operational Efficiency

The ability of RS Gas to deliver services in the most cost-effective manner while



maintaining high-quality standards. It includes reduced wait times, minimized errors, and streamlined refueling operations due to automation.

Technology Acceptance Model (TAM)

A theoretical framework that explains how users come to accept and use new technology, focusing on two main factors: *Perceived Usefulness* and *Perceived Ease of Use*. This model guides the assessment of consumer readiness to adopt the automated refueling system.

Perceived Usefulness (PU)

The degree to which a person believes that using a particular system would improve their performance—in this context, referring to how much users believe the automated system enhances their refueling experience.

Perceived Ease of Use (PEOU)

The degree to which a person believes that using the system would be free of effort. This term evaluates how intuitive and user-friendly the automated refueling interface is for customers.

Contactless Payment

A method of payment that allows transactions without physical contact between the payment device and the machine, often using NFC technology, QR codes, or RFID tags.



Demographic Profile

A statistical overview of the participants involved in the study, including characteristics such as age, gender, vehicle type, payment preference, and frequency of gas refueling.

Transaction Speed

The amount of time it takes to complete a fuel purchase using the automated system, from initiating the refueling process to completing the payment.

User Trust

The confidence users have in the system's reliability, security, and effectiveness especially in handling their payment data and providing accurate fuel delivery.

Fuel Service Industry

An industry sector that provides petroleum-based products such as gasoline and diesel to consumers, primarily through fuel stations.